Customer booking prediction: geography is the strongest signal

CUSTOMER ANALYTICS

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Random Forest model on 49,281 British Airways booking enquiries | 5-fold cross-validation + holdout evaluation

0.79
ROC-AUC
Holdout | CV 0.795 ± 0.003

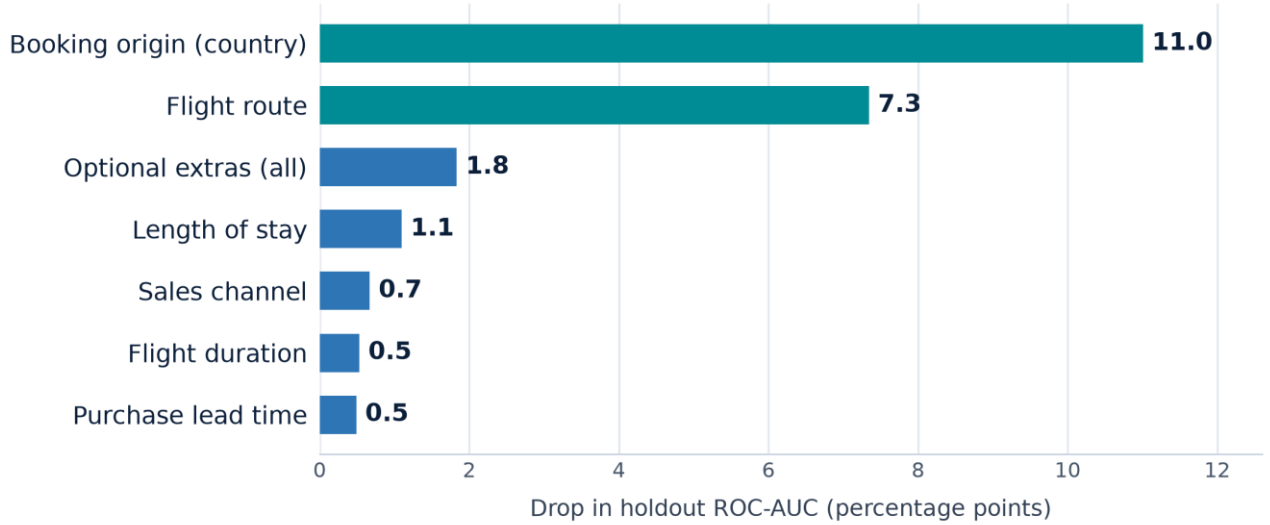
0.37
PR-AUC
2.4x the 15% base rate

2.9x
LIFT AT TOP 10%
vs contacting a random 10%

29%
BOOKINGS CAPTURED
by targeting the top 10%

What drives a booking?

Drop in holdout ROC-AUC when each driver is shuffled. Other features add <0.1 pts each.



WHAT THE MODEL TELLS US

1. **Market and route drive ~80% of measured importance.**
   Conversion is ~5% in Australia/New Zealand vs ~35% in Malaysia. The score mostly separates markets before it separates individuals.

2. **Intent signals add usable, but smaller, lift.**
   All three extras: 18.6% vs 10.7% with none. Short breaks (4-7 days): 19.5% vs 10.1% for 15-30 days. Internet 15.5% vs Mobile 11.0%.

3. **A useful ranking, not a guarantee.**
   PR-AUC 0.37 vs 0.15 base rate; at the 0.5 threshold the model finds 61% of bookers with 34% precision.

Accuracy is 77% - below the 85% 'never books' baseline. Precision at the top of the ranking is the metric that matters.

RECOMMENDATION

Rank enquiries and target the top 10-20% for 2.5-2.9x lift; A/B test incremental bookings before scaling.
Localise offers by origin/route and short breaks; retrain quarterly with price and seasonality data.

Holdout (n=9,857): ROC-AUC 0.788 | PR-AUC 0.367 | recall 0.61 | precision 0.34 at 0.50. CV: ROC-AUC 0.795 ± 0.003.

Data: 50,000 rows, 719 exact duplicates removed. 80/20 stratified split; 5-fold CV; 600-tree Random Forest; out-of-fold target encoding for route and origin. Importance = grouped drop in ROC-AUC when a driver is shuffled.